CSA1701 - ARTIFICIAL INTELLIGENCE

3.Question: Write a Python program to solve the Water Jug problem.

Aim: To measure exactly 2 litres of water using a 4-litre jug and a 3-litre jug.

Algorithm:

1. Start with both jugs empty.

2. Represent the state as the amount of water in each jug.

3. Generate states by filling, emptying, or pouring water between jugs.

4. Use BFS to explore the states.

5. Stop when one jug contains exactly 2 litres.

Program:

2 python.py - C:/Users/praji/AppData/Local/Programs/Python/Python314/2 python.py (3.14.8)

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```
from collections import deque

CAP_A, CAP_B = 4, 3
start = (0, 0)
goal = 2

def next_states(a, b):
    return [
        (CAP_A, b),
        (a, CAP_B),
        (0, b),
        (a, 0),
        (a - min(a, CAP_B - b),
         b + min(a, CAP_B - b)),
        (a + min(b, CAP_A - a),
         b - min(b, CAP_A - a))
    ]

queue = deque([(start, [start])])
visited = {start}

while queue:
    (a, b), path = queue.popleft()

    if a == goal or b == goal:
        print("Solution path:")
        for state in path:
            print(state)
        break

    for state in next_states(a, b):
        if state not in visited:
            visited.add(state)
            queue.append((state, path + [state]))
```

IDLE Shell 3.14.8

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```
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/praji/AppData/Local/Programs/Python/Python314/3.python.py
Solution path:
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
>>>
```

Result: Thus, exactly 2 litres of water were measured using the Water Jug problem.